

Three Unwritten Papers in Symmetric Cryptography

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Generic Attacks and Proofs in
Symmetric Cryptography (GAPS 2)

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*Thanks to Elena & Stefan for the
kind invitation to come give a talk.*

Please come sit in
the front 😊



Today

People usually give talks on papers they've written.

I thought it would be fun to talk about some papers I *didn't* write—but had thought to, long ago.

1. Garbled Circuits in the Lockbox Model
2. The Pervasive Hardness Conjecture for Blockciphers
3. On Obfuscation with Random Oracles

Provisos:

- Long ago $\Rightarrow$ largely forgotten
- None fully baked. Maybe naïve
- Only 1.5 of 3 is in an area I worked on

The realized **and envisaged** garbling papers

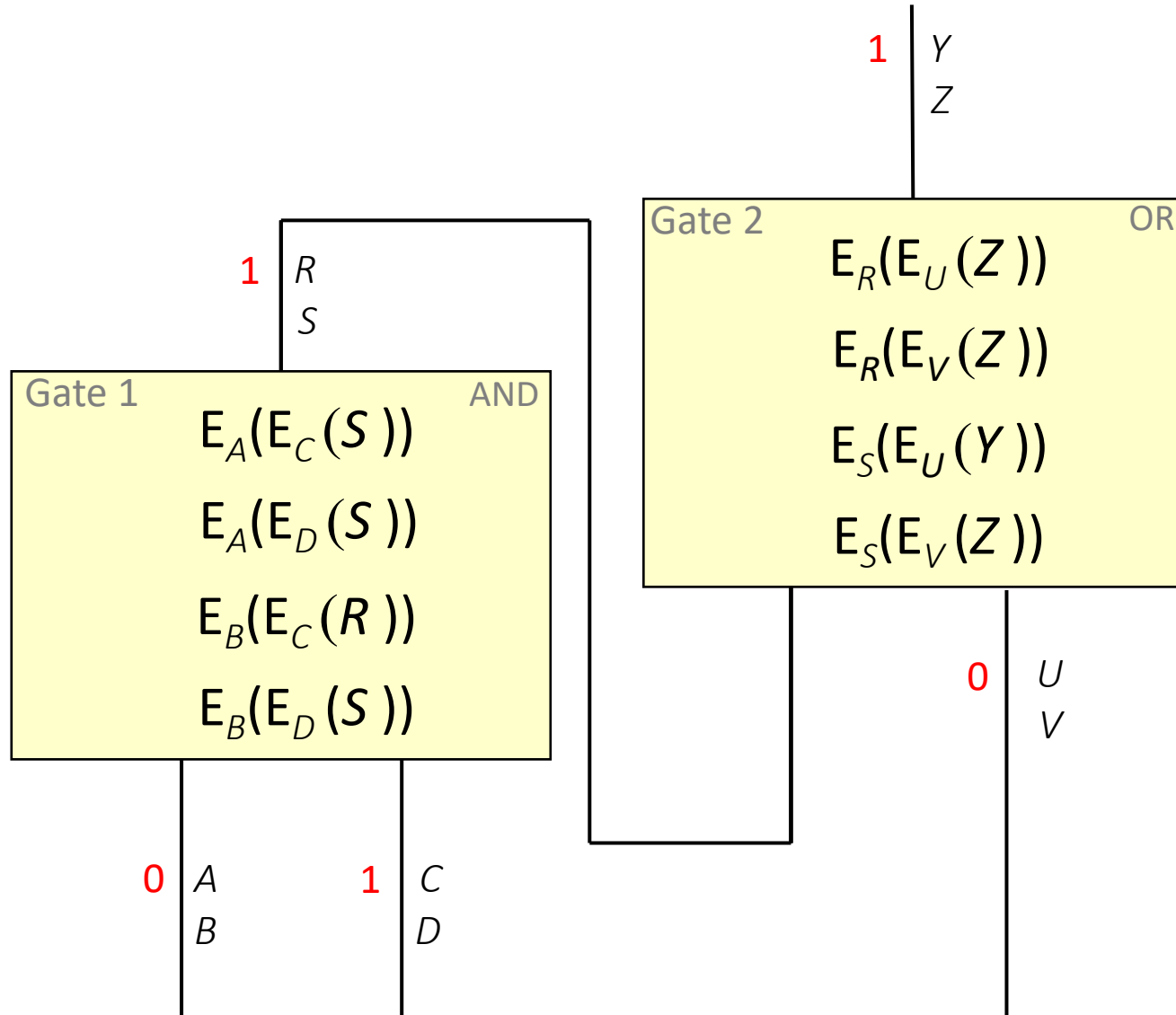
1. **Foundations of Garbled Circuits**
[Bellare, Hoang, Rogaway 2012]
2. **Adaptively Secure Garbling with Applications to One-Time Programs and Secure Outsourcing**
[Bellare, Hoang, Rogaway 2012]
3. **Efficient Garbling from a Fixed-Key Blockcipher**
[Bellare, Hoang, Keelveedhi, Rogaway 2013]
4. **Garbled Circuits in the Lockbox Model**
[Nobody, never]

Paper 1

Sketched in a 2012 talk;
conceived ~1990

One account of “Yao’s Protocol” [1982/1986?]

[Lindell, Pinkas 2004/09] *A Proof of Security of Yao’s Protocol for Two-Party Computation*



Use of double encryption for this seems to date to Goldreich 2001: Cryptography and Cryptographic Protocols

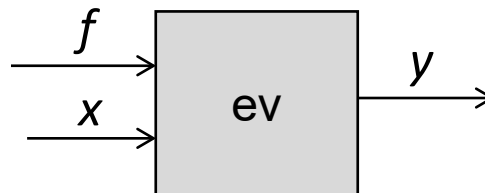
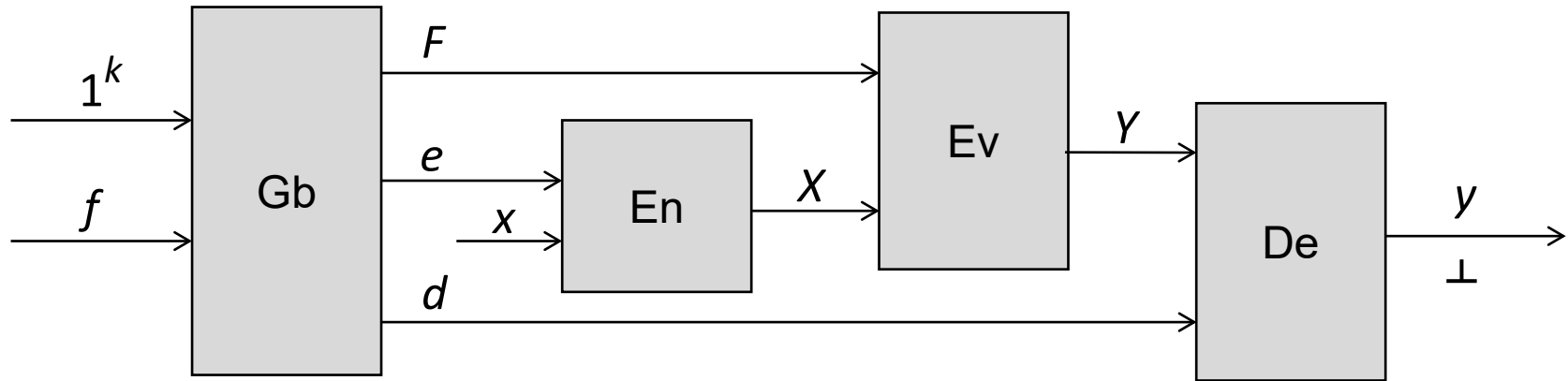
Creating Foundations

Syntax from [BHR2012]

prob.

det.

A **garbling scheme** is a 5-tuple $G = (Gb, En, De, Ev, ev)$



$(\forall f, x, k)$, if
 $(F, e, d) \leftarrow Gb(1^k, f)$,
 $X \leftarrow En(e, x)$,
 $Y \leftarrow Ev(F, X)$,
 $y \leftarrow De(d, Y)$ then
 $y = ev(f, x)$

Correctness

There's another approach to interrogating the foundations of garbled circuits

Physical, dual-lock lockboxes

Dual Lock STEELMASTER® Key Cabinet
10 Key Capacity – Keyed Differently

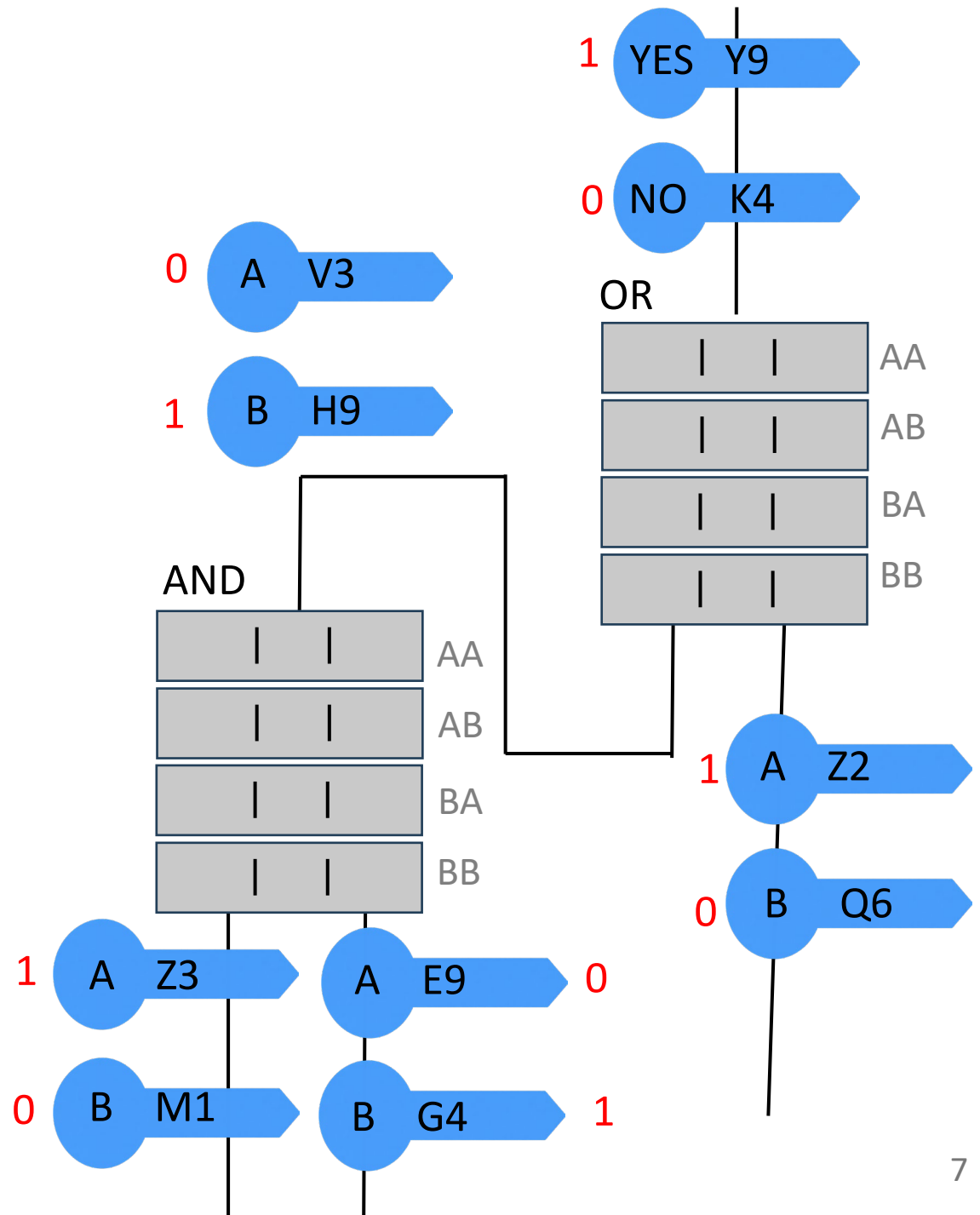


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Garbled circuits from physical lock boxes

“The ... construction can be described **metaphorically** using “locked boxes”. The ... idea ... is that every wire is allocated two padlock keys; one key is associated with the bit zero, and the other with the bit one. Then, for each gate four doubly-locked boxes are provided, The four boxes are locked so that each pair of keys ... opens exactly one box. [In] each box a single key relating to the output wire of the gate is stored.”
[Lindell, Pinkas 2009]



Lockbox Model

A set $\mathbb{T}$ of **tokens** (or **capabilities**), any of which you might **possess**.

A set $\mathcal{R}$ of **rules** $(A, B) \mapsto C$.

Call $\text{Try}(A, B)$ to recover all tokens T s.t. $(A, B) \mapsto T \in \mathcal{R}$.

Once you possess a token T you can read its **label** $\text{Lbl}(T)$.

$(H9, Q6) \mapsto Y9$

$(H9, Z2) \mapsto Y9$

$(M1, E9) \mapsto V3$

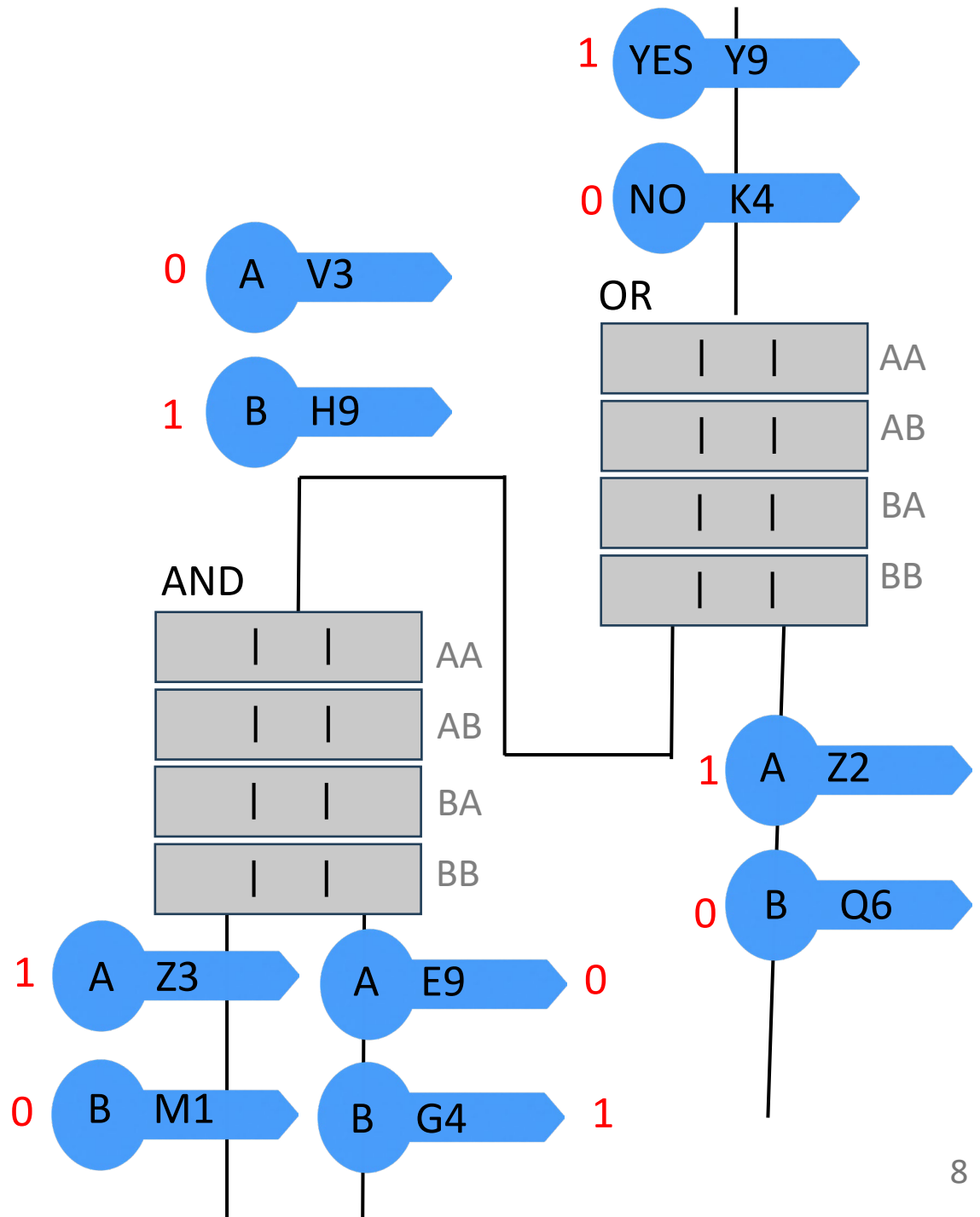
$(M1, G4) \mapsto V3$

$(V3, Z2) \mapsto Y9$

$(V3, Q6) \mapsto K4$

$(Z3, E9) \mapsto V3$

$(Z3, G4) \mapsto H9$



The starting point is simply to *take the model seriously*.

Build a minimalist, purpose-built model. Investigate instantiations of a lockbox that give rise to a provably-secure garbling scheme. Are there other physically inspired models for garbling or MPC? Etc.

[Li, Micciancio 2018] *Symbolic security of garbled circuits*. Starts with an [Abadi, Rogaway]-like algebra for encryption and extends it with stuff like PRGs and a random-swap operator to get a model rich enough to describe a specific garbling scheme.

For me: a garbled circuit is not conceptually encryption. It deserves its own dedicated formal treatment, one that directly models lockboxes.

General suggestion: For work that is not rigidly follow-on work, invent what you need **specifically** for your task. You owe no loyalty to past definitions, frameworks, or theorems—including your own. It's all disposable. *Your treatment of a topic shouldn't be shaped by inherited machinery, but by the intrinsic ontology of that topic.*

Paper 2:

The Pervasive-Hardness Conjecture for Blockciphers

Memo to the Amateur Cipher Designer

Anyone, from the most clueless amateur to the best cryptographer, can create an algorithm that he himself can't break.

Bruce Schneier, 1998 *Memo to the Amateur Cipher Designer, 1998*

Creating hardness is hard

Most attempts to do it will fail

Paper 2:

The Pervasive-Hardness Conjecture for Blockciphers

Memo to the Amateur Cipher Designer

Anyone, from the most clueless amateur to the best cryptographer, can create an algorithm that ~~he himself can't~~ break.

nobody can

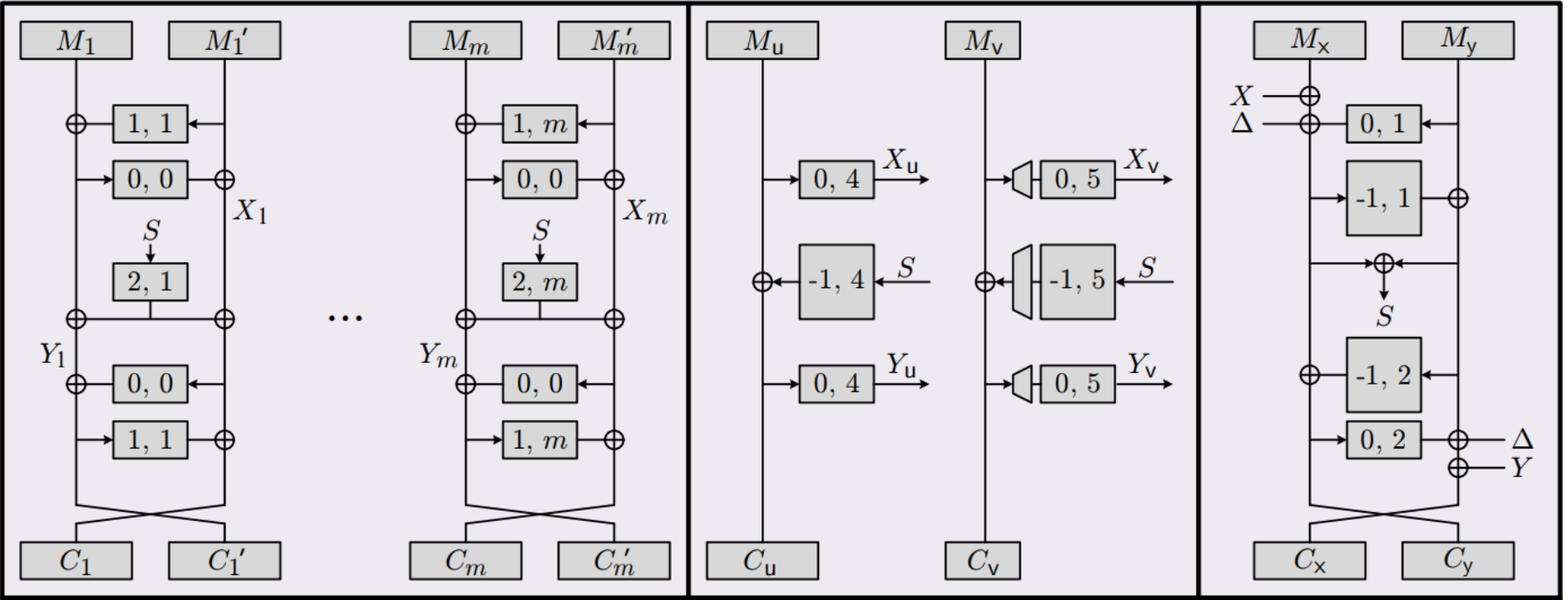
Bruce Schneier, 1998 *Memo to the Amateur Cipher Designer, 1998*

Creating hardness is ~~hard~~ **easy**

Most attempts to do it will ~~fail~~ **succeed** (as long as you're not stingy
with resource use)

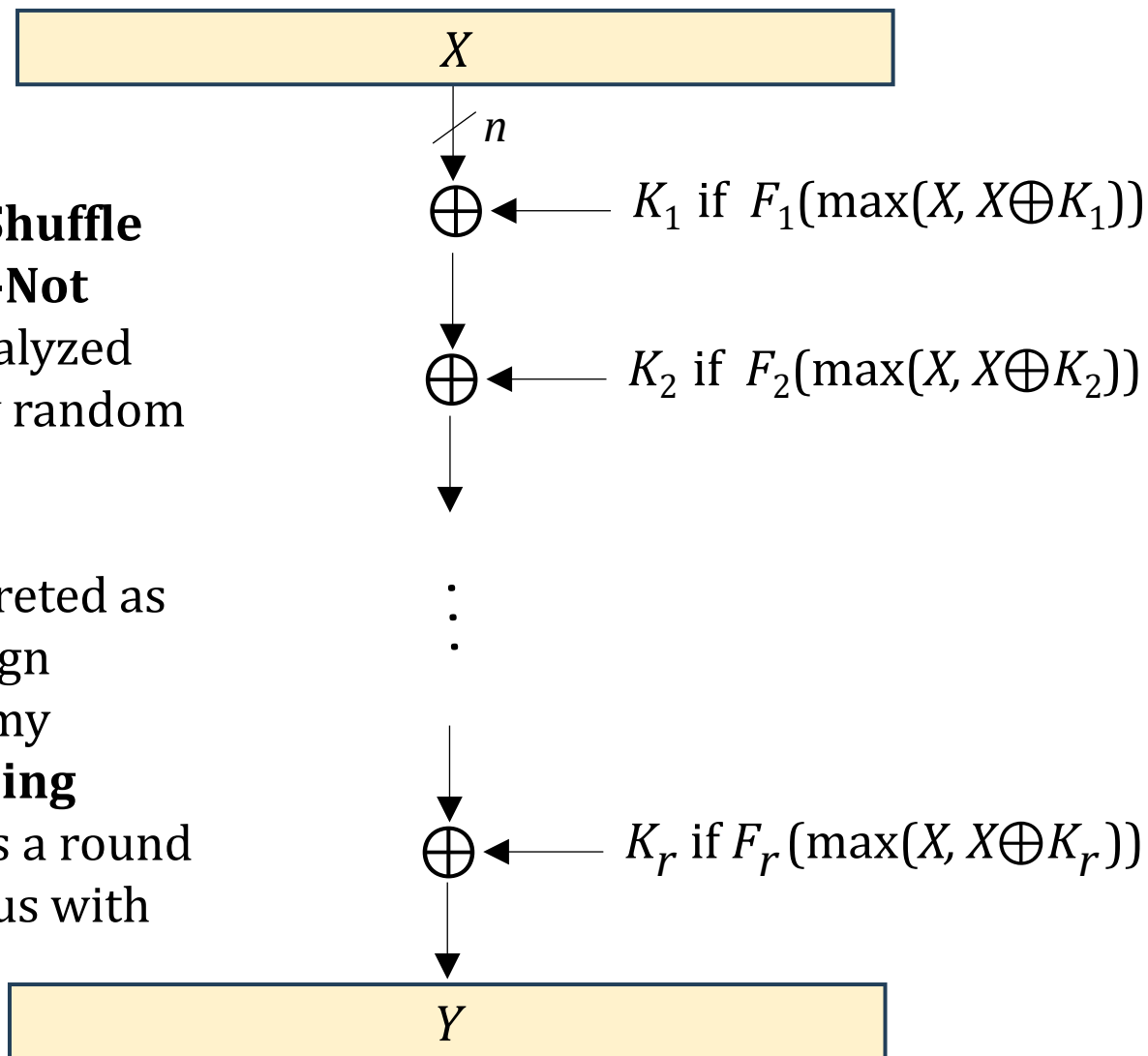
The design difficulty is efficiency — not creating hardness

I did two confusion/diffusion designs: **SEAL** (1996/1998) and **AEZ** (2015). They were complex, fiddly, and security felt dicey. But that was **only** because of aggressive efficiency aims. If I had **just** wanted a secure PRP? *Seems trivial.*



I also analyzed the **Thorp Shuffle** and invented this **Swap-or-Not** cipher with a colleague. Analyzed them assuming a uniformly random round function F_i

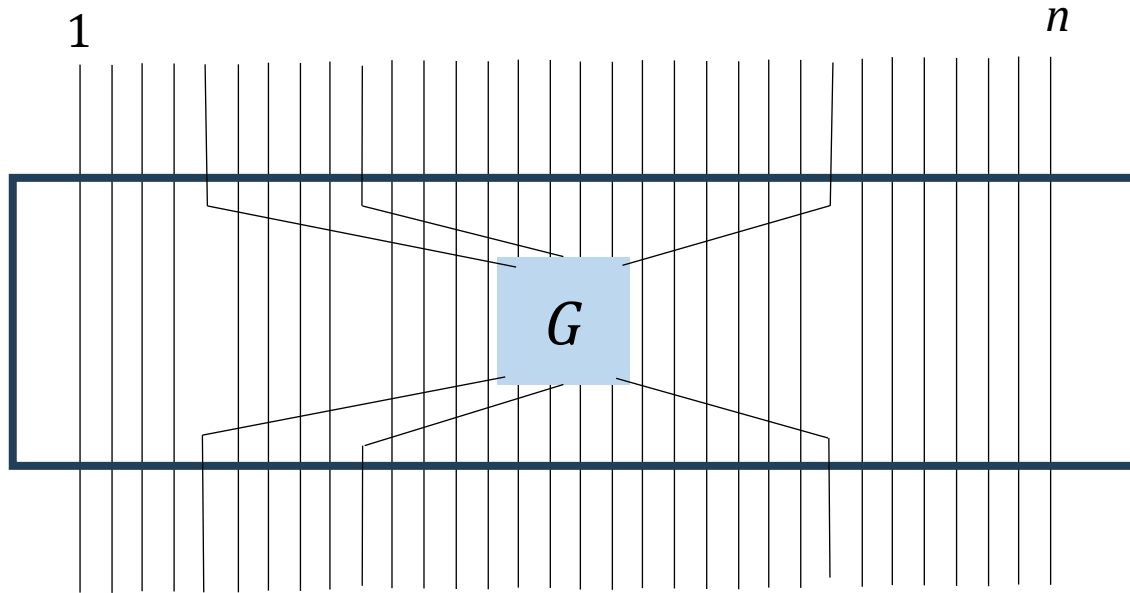
But both can also be interpreted as pragmatic blockcipher design templates. In that context, my expectation was that **anything reasonable** should work as a round function F_i if you're generous with the number of rounds.



Swap-or-Not [Hoang, Morris, Rogaway 2018]

Simple round functions may suffice

The Gowers Cipher



[Gowers 1996] *An almost m -wise independent random permutation of the cube*

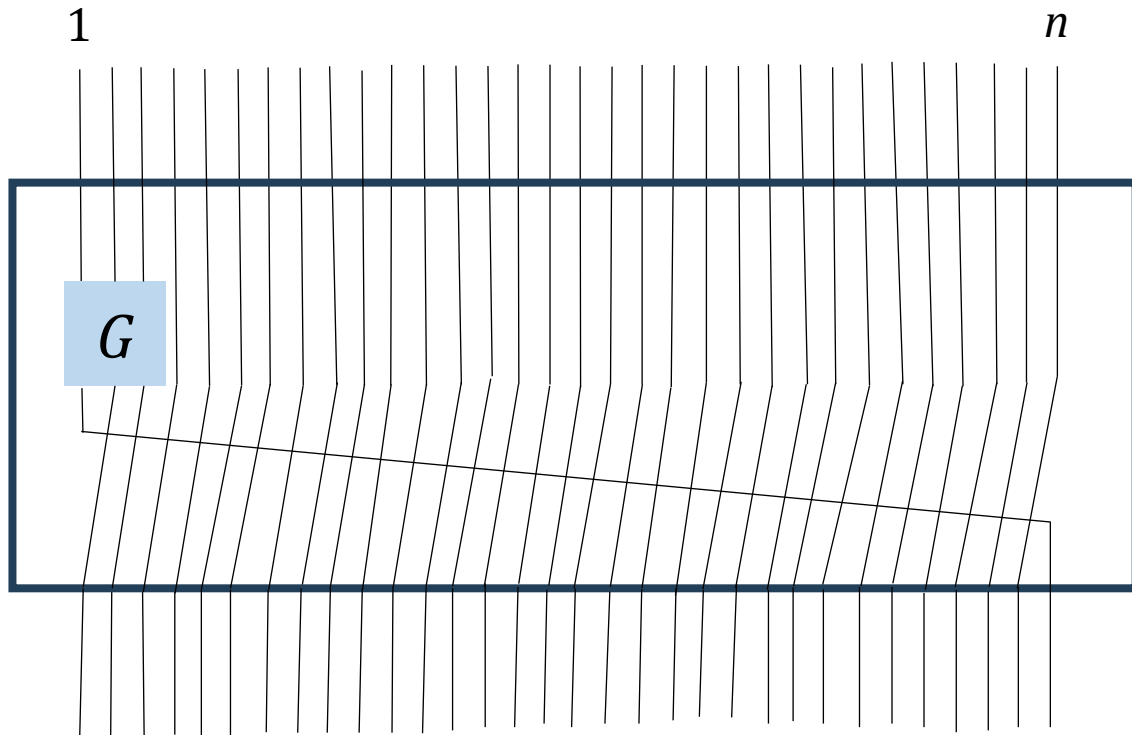
Each round key names three wires in $[1..n]$ and a permutation $G \in S_3$ to apply to those wires.

Gowers showed that if you iterate this enough rounds, you get k -wise independence: $r_k \in \tilde{O}(n^8 k \lg(1/\epsilon))$. This has been repeatedly improved (sometimes with a tweaked construction) [Hoory, Magen, Myers, Rackoff 2004], [Brotsky, Hoory 2005], [He, O'Donnell 2024], [Gretta, He, Pelecanos 2024/25], the last reaching $r_k \in O(nk \lg(1/\epsilon))$.

What might the Gowers result mean concretely for $n=128$? Hard to know. Maybe nothing.

Gowers suggests that the construction above might in fact be a PRP. If so, surely one of the simplest blockcipher suggestions ever.

Even simpler: how about 1BIT[n, r, G_0, G_1]



A **1-bit round key** selects whether you apply $G_0 \in S_8$ or $G_1 \in S_8$ for **fixed** G_0 and G_1 . Then ROL.

I suggest that even 1BIT is a good PRP for most (G_0, G_1) and a large yet reasonable number of rounds r .

Maybe a good test piece:
which (G_0, G_1) work?

Even: $G_0 = \text{id}$ and

or $G_1(x, y, z) = (x \oplus (y \vee \bar{z}), y, z)$ “G11”

$G_1(x, y, z) = (x \oplus (\bar{y} \vee z), y, z)$ “G13”

plausibly work (the only width-2,
3-bit gates that aren’t obviously bad)

algorithm $E(K, X)$

for $i = 1$ **to** r **do**

$X[1] \wedge= K[i] \ \& \ (\sim X[2] \mid X[3])$

$X = \text{ROL}(X)$

return X

Pervasive hardness conjecture: It's *easy* to make a secure blockcipher: do *anything reasonable* for a round function, for a sufficient but reasonable number of independently keyed rounds.

- **Engineering claim:** That will work.
- **Mathematical claim:** There should be a weak, understandable, and often verifiable condition on the round function R under which iteration gives a good PRP.

A proposed condition: **progressive mixing**

Let r_k = the **k -mixing time**: the minimum number of rounds until any k distinct points map to nearly uniform distinct points. (Depends on a distance ϵ , which I'm omitting.) The **mixing profile** is the sequence $r_1, r_2, r_3 \dots$.

Progressive mixing = a good mixing profile.

Conjectures

1. Blockcipher round functions that seem “reasonable” are progressive mixing.
2. A progressively mixing round function will give you a secure blockcipher if you iterate a reasonable number of rounds.
3. A better mixing profile suggests you need fewer rounds for PRP security.

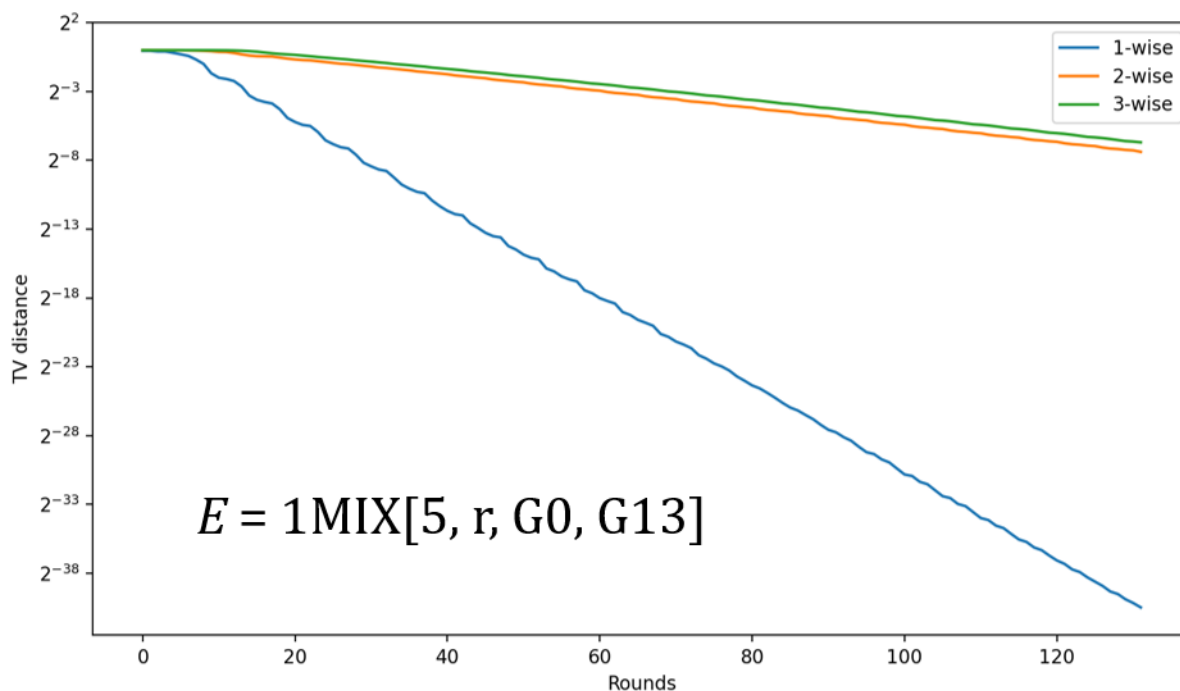
Challenge: Compute the first few terms of the mixing profile—concrete numbers—for real or minimalistic constructions.

Some mixing times for 1BIT instances

Exploratory computation by ChatGPT — not independently verified

n	r_1	r_2	r_3	r_4	$\varepsilon = 0.01$ TVD
3	15	47	52	∞	
4	24	101	110	118	
5	25	120	131	137?	
6	33	181	199		

```
algorithm  $E(K, X)$ 
for  $i = 1$  to  $r$  do
     $X[1] \wedge= K[i] \ \& \ (\sim X[2] \mid X[3])$ 
     $X = \text{ROL}(X)$ 
return  $X$ 
```



[HMMR04] hoped that small r_4 would be enough

[Hoory, Magen, Myers, Rackoff 2004] *Simple Permutations Mix Well.*

Conjectured that once a “simple” blockcipher had mixed enough to achieve almost 4-wise independence, you *already* have a secure PRP.

- 2-wise independence already subsumes much cryptanalysis
- 3-mixing is not enough (affine maps on $\{0,1\}^n$ are 3-wise independent)
- 4-mixing implies that the group generated by the blockcipher’s round function is huge: either S_N or A_N . CFSG establishes that all 4-transitive subgroups of S_N contain A_N , with the exception of the four Mathieu groups $M_{11}, M_{12}, M_{23}, M_{24}$, none of which even have a power-of-2 degree.

[Dujmovic, Pelecanos, Tessaro 2026] *When Simple Permutations Mix Poorly: Limited Independence Does Not Imply Pseudorandomness.* Disproved HMMR04 conjecture.

Their construction is 4-wise independent after $n-2$ rounds, but not 5-wise independent. But after a few more rounds, the 5-point dependency would seem to fall apart. Maybe they just need to mix some additional rounds? I doubt it’s enough, in general: I suspect one can make weird counterexamples that retain 5-wise near-dependency for a long time after 4-wise independence is reached.

Paper 3: On Obfuscation with Random Oracles

VBB obfuscation

[Barak, Goldreich, Impagliazzo, Rudich, Sahai, Vadhan, Yang 2001]

On the (Im)possibility of Obfuscating Programs

That which an adversary can output given an obfuscated program $\mathcal{O}(P)$ the adversary can output just as well with only oracle access to P .

In sketch:

$$(\forall A)(\exists S)(\forall P) \quad A(\mathcal{O}(P)) \approx S^P(1^{|P|})$$

I like this notion, but my first instinct is to simplify:

Forget indistinguishability. You have only a RO. Statistical security. Go.

Also: Not fond of $(\forall A)(\exists S)$ notions for concrete-security results, because you must worry about how S depends on A , and what are its resources.

Papers doing information-theoretic VBB obfuscation in the ROM

[Lynn, Prabhakaran, Sahai 2004]

*Positive Results and Techniques
for Obfuscation*

Point functions: $x=a$? Small sets: $x \in A$?

General-output point functions: $a_i \mapsto b_i$

Secret labeled graph traversal.

Regular expression pattern matching.

[Banerjee–Galbraith–Russello 2023]

Obfuscating Evasive Decision Trees

Intervals. Evasive decision-tree classifiers
with hidden thresholds.

Paper-3 agenda

- a) Provide a concrete, indistinguishability-style definition for VBB obfuscation.
- b) Add in the RO.
- c) Relate it to the simulator-based definitions.
- d) Rework all past results, with concrete bounds, for functions that admit information-theoretic VBB obfuscation in the ROM.
No complexity assumptions; the ROM is the only source of hardness.
- e) Compare with ROM iO. Same deal: statistical indistinguishability, no hardness assumption.
- f) Characterize exactly *what* functionalities admit information-theoretic obfuscation in the ROM.

3a) A concrete, indistinguishability-style definition for VBB obfuscation

Obfuscator: probabilistically maps $\mathcal{O}: f \mapsto g$ where $f = g$

advantage
using **code**

$$p_A = 2 \Pr[(f_0, f_1, s) \leftarrow A(\text{find}); \\ g_0 \leftarrow \mathcal{O}(f_0); g_1 \leftarrow \mathcal{O}(f_1); b \leftarrow \{0,1\}; \\ b' \leftarrow A(g_b, g_{1-b}, s): b = b'] - 1$$

optimal
advantage
using
black boxes

$$\delta_{A,q} = \max_{q\text{-query } D} \left[2 \Pr[(f_0, f_1, s) \leftarrow A(\text{find}); b \leftarrow \{0,1\}; \\ b' \leftarrow D^{f_b, f_{1-b}}(s): b = b'] - 1 \right]$$

$$\text{Adv}_{\mathcal{O}}(A, q) = \max(0, p_A - \delta_{A,q})$$

Special case, iO: if $f_0 = f_1$ (with equal-length descriptions) then $\delta_{A,q} = 0$

Concluding remarks

We looked at:

1. Garbled Circuits in the Lockbox Model
2. The Pervasive Hardness Conjecture for Blockciphers
3. On Obfuscation with Random Oracles

Anything in common?

Perhaps exemplify some sort of common **taste**.

Personal taste matters.

My taste: A problem that was never recognized as a problem. A paper that will *start* something. A nice subset of {definitional, overlooked, practical, simple, timeless}.

Your taste is probably different. Whatever it is: embrace it. Refine it. Research goes best when the problems you work reflect an *identifiable personal taste*.

Three Unwritten Papers in Symmetric Cryptography

Abstract. People usually give talks about papers they've written. I thought it would be fun to turn that around and talk about papers that I *didn't* write. I'll reach into the distant recesses of memory to identify three problems in symmetric cryptography that I once thought would make for good papers, but that I never pursued. They failed to materialize not because I got stuck, but just because, you know, we don't do everything we think of doing. I won't tell you here what the three non-papers are about, because that would spoil the surprise.